

Aakash Thakkar

Full Stack Engineer | Industrial Robotics & HMI

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SUMMARY

Full stack engineer in industrial robotics, working where the operator interface meets the machine. Main frontend developer on an HMI builder platform whose output runs on **60+ deployed palletizing cells** at customers including WAGO, Albéa and Arvato. Comfortable at both ends of that stack: React, TypeScript and Three.js in the browser, Python and FastAPI behind it, Modbus TCP and OPC UA out to the PLC and cobots. MSc Computer Science, Universität Paderborn.

EXPERIENCE

• Full Stack Developer

10.2022 – Present | Paderborn, Germany

Unchained Robotics GmbH 

HMI Builder Platform

TypeScript, React, Three.js, Python, FastAPI, gRPC/Protobuf, WebSockets, Kubernetes, Docker, GitHub Actions

- Main developer of the feature set of a web-based drag-and-drop HMI builder. Around 25 application programmers use it to configure operator HMIs, which run on 60+ deployed robot cells in food, automotive and part-handling production (WAGO, Albéa, Arvato).
- Set the technical direction for the HMI layer, translating product requirements into the widget data model and the PLC interaction contract used by ~25 application programmers.
- Built most of the widget library. The pallet preview renders a full pallet in 3D, layer by layer, in Three.js. The pallet status widget shows layers in 2D with per-box active and inactive indicators. The image carousel pulls images from file storage or by name from the PLC.
- Wrote the FastAPI backend holding HMI configuration (products, pallet schemas, taught positions) for both the builder and the runtime HMI.
- Maintains the pnpm monorepo and owns releases: PR review, a major release every 3–4 months, minor releases roughly monthly, cloud editor deployed on Kubernetes through GitHub Actions.
- Runs sprint planning and capacity planning together with the team lead, in a team of three.
- Wrote Python simulators for PLC signals so widgets can be built and demonstrated without robot hardware.

Sequencing and Multi-Pick Logic

TypeScript, React, computational geometry

- Rewrote the automatic sequence generator for pallet schemas. The old one produced sequences that collided, so operators had to check every one by hand. The replacement uses a corridor-sweep method and runs client-side.
- Added sequence validation with configurable sweep clearance, so hand-written sequences are checked for collisions before they reach a cell.
- Built the multi-pick and multi-place feasibility check. It takes gripper and product dimensions, feeding orientation and the boxes the operator selected, works out whether those boxes fit the gripper together, and then whether they can go down in one move or have to be split into sub-moves (13.1, 13.2) while staying a single pick.
- Surfaced grouping in the UI and in progress handling, so operators can select a group when starting or resuming a job.

Robot, PLC and Vision Integration

Doosan, Universal Robots, Modbus TCP, OPC UA, Python, Siemens coupler

- Programs and commissions Doosan and Universal Robots cobots for palletizing and depalletizing: teach pendant, joint moves scripted in Python, Modbus register reads and writes.
- Works daily against an in-house PLC platform written in Python, step-chain logic over a Siemens coupler, speaking Modbus TCP and OPC UA.
- Second-level support for the PLC programmers, tracing signal and communication faults across the HMI, PLC and robot layers, including faults that turn out to sit outside the HMI.
- Built a web HMI for an automated inspection cell in Python, Django, Docker and WebSockets. Operators teach inspection points, capture images and text, and edit and export reports. Integrated the Inspekto S70 vision system.

• Front End Developer (20h/week)

03.2022 – 08.2022 | Paderborn, Germany

Unchained Robotics GmbH 

- Migrated an e-commerce platform from WordPress to NextJS, working with GraphQL against a Saleor backend.
- Added features to an existing web-based HMI, including the pallet schema editor, and worked on a Python updater thread pushing message-system updates to the database and HMI over WebSockets.

• **Full Stack Developer (20h/week)**

10.2020 – 12.2021 | Paderborn, Germany

Floccess GmbH

- Sole developer of a heat exchanger calculation web application (AngularJS, Django, SQL, Nginx), from build through testing to deployment.
- Added **ThreeJs** visualisations of exchanger vessels and interactive forms graphing calculated temperature and volume variations.

• **Student Research Assistant (20h/week)**

10.2019 – 09.2020 | Paderborn, Germany

Universität Paderborn

• **Systems Engineer**

2016 – 2019 | Delhi, India

Tata Consultancy Services Pvt. Ltd. 

- Built a Java tool called Legacy Adapter that pulls around **a million JSON messages a day** off a remote server, parses them and writes the extracted attributes into database tables, with automated tests flagging discrepancies against the source.
- Built mobile-compatible UI dashboards for retail data analysis in AngularJS and Node.js, and supported UNIX deployments with Git, SVN, SFTP and shell scripting.

EDUCATION

• **Master of Science - Computer Science**

2019 – 2024 | Paderborn, Germany

Universität Paderborn

- Master Thesis: Fuzzing with Static Analysis - Java, Maven, Bazel, Jazzer, JQF
Combined coverage-guided fuzzers with static data flow analysis, using the analysis to generate initial corpora and validate inputs.
- Linked Data to Natural Language (LD2NL) - Project Group
Built a summary generator for linked data (**Knowledge Graphs**) with a React frontend over a Java REST API, ordering output by node prominence.

• **Bachelor of Technology - Computer Science & Engineering**

2012 – 2016 | Vellore, India

VIT University

- Bachelor Project: Automatic Traffic Jam Detection
Wrote an image recognition program in Python and OpenCV that detects and quantifies traffic congestion in real time, then sets traffic signal states from current conditions.

LANGUAGES

English — C1 • **Deutsch** — B1

SKILLS

Robotics — Doosan and Universal Robots cobots, Modbus TCP, OPC UA, PLC step-chain logic, palletizing and depalletizing, machine vision (Inspekto)

Frontend — TypeScript, React, NextJS, Three.js, Tailwind

Backend — Python (FastAPI, Django), gRPC/Protobuf, WebSockets, REST, GraphQL

Languages — Python, TypeScript/JavaScript, Java

DevOps — Docker, Kubernetes, GitHub Actions, CI/CD, Git

Data — PostgreSQL, MongoDB

Process — Sprint and capacity planning, release management, PR review, Jira, Agile